ICS EXE 4

2026.03.27

Problem 1

1. 1/2

2. No, (Reordering line 8 and 7 does not reduce miss rate)

3. B. Increasing the block size enables filling more elements in each cache line, thus reduce the miss rate.

Problem 2

1 [1] 59 [2] 2 [3] 3

2 If the high-order bits are used as an index, then some contiguous memory blocks will map to the same cache set. If the program scans an array sequentially, then the cache can only hold them in one set, cannot make full use of the whole cache. Use middle-bit indexing can map adjacent blocks to different cache sets.

3

Order	Operation	Hit/Miss	Byte Returned
1	Read 0x401368	Miss	--
2	Read 0x8036a5	Hit	0x45
3	Write 0x8036b3	Miss	--
4	Write 0x4014aa	Miss	--
5	Write 0x40139b	Hit	--
6	Read 0x8026cf	Miss	0xda

4 5. After 6 operations, the final state of the cache has 4 cachelines with the dirty bit set to 1. Including the one eviction that occurred during the process, the total count is 5.

5 All miss.

6 Reorder line 7 and line 8.

7 ABC